

Sanjeet Kumar Nayak

📍 Hyderabad, India | 📞 +91 8908739585 | ✉️ sanjeetkumariit@gmail.com | 🌐 in/sanjeet-kumar-nayak

SUMMARY

Mechanical Engineer with 11 years of experience in HVAC design, building energy modelling and building performance analysis. Experienced in leading engineering teams, developing ASHRAE-compliant energy models, and delivering projects across commercial, institutional and utility sectors. Worked on projects in the US, Middle East, Africa and Asia, with hands-on experience in HVAC systems, building controls and energy performance analysis

WORK EXPERIENCE

Engineering Manager – Energy Modelling

Alliantgroup - Hyderabad, India

March 2021 – Present

Lead an engineering team delivering ASHRAE 90.1 energy models for Section 179D energy efficiency projects across commercial buildings in the United States.

Responsibilities

- Lead a team of 15 engineers by assigning tasks, reviewing technical deliverables and monitoring project schedules, improving delivery quality and consistency.
- Develop ASHRAE 90.1 baseline and proposed energy models using construction drawings, mechanical schedules and design specifications.
- Review HVAC systems including AHUs, RTUs, chillers, boilers and heat pumps to ensure accurate representation in energy simulations.
- Analyse energy model outputs to verify compliance with ASHRAE performance requirements and identify modelling issues.
- Collaborate with engineering and tax teams to resolve technical questions and maintain project schedules.
- Establish modelling standards and quality assurance procedures that reduced rework and improved consistency.
- Mentor junior engineers through technical reviews and structured feedback sessions.

Key Achievements

- Delivered energy modelling for 450+ commercial buildings.
- Managed projects generating over US\$2 million in quarterly revenue.
- Developed an automated baseline model generation process that significantly reduced manual modelling effort.
- Led quality assurance reviews for 300+ buildings.
- Authored standard operating procedures for ASHRAE energy modelling adopted across the team

Assistant Engineering Manager – Mechanical (HVAC)

L&T Construction - Chennai, India

June 2015 - March 2021

Designed HVAC systems for power plants and infrastructure projects across the Middle East and Africa.

Responsibilities

- Designed chilled water and DX HVAC systems using Carrier HAP and ASHRAE design standards for power plants and infrastructure projects across the Middle East and Africa.
- Performed cooling and heating load calculations and selected HVAC equipment.
- Prepared duct layouts, ventilation calculations and mechanical design documentation.
- Reviewed vendor submissions and ensured compliance with project specifications.

- Coordinated with architectural, structural and electrical disciplines to resolve design clashes before construction.
- Supported technical discussions with clients, consultants and equipment suppliers.
- Reviewed construction drawings and engineering documentation throughout project execution.

Key Achievements

- Delivered engineering packages within planned schedules and allocated man-hours.
- Secured client approval for engineering submissions with minimal review comments.
- Supported successful delivery of international infrastructure projects across six countries

EDUCATION

Bachelor of Technology (Mechanical Engineering)

Indian Institute of Technology (IIT) Bhubaneswar 2015

SKILLS

Industry Knowledge: Building Energy Modelling, HVAC Design, HVAC Controls & BMS, Building Performance Analysis, LEED Energy Modelling, Energy Auditing, Measurement & Verification (Exposure), ASHRAE Standards, HVAC Load Calculations, Mechanical Design, Project Management, Team Leadership, Client Coordination

Technical Skills: eQUEST 3.65, DesignBuilder 7.3.1, Carrier HAP 6.2, AutoCAD 2025, Microsoft Excel, Microsoft Office, ASHRAE 90.1, ASHRAE 62.1